

YITIAN WANG

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SUMMARY

Ph.D. scholar in Electrical Engineering specializing in thermal transport and lattice dynamics in functional oxides and solid-state ionic conductors. Experienced in crystal and thin-film growth, vacuum deposition, material characterization, and data-driven modeling using Python and MATLAB. Published 5 first-author works in leading journals and skilled at collaborating with national labs and cross-disciplinary device and materials teams. Additional teaching assistant experience during Ph.D. studies.

EDUCATION

University of California, Riverside, CA

Mar. 2021 - Jun. 2026 (expected)

Ph.D. in Electrical and Computer Engineering

Awards: Dissertation Completion Fellowship Award;

MRS 2023 Spring Highly Commended Student Talk Award

Columbia University in the City of New York, NY

Aug. 2019 - Feb. 2021

Master of Science, Material Science and Engineering

University of California, Berkeley, CA

Jul. 2017 - Aug. 2017

Summer Session, Physics

Beijing Normal University, CN

Sep. 2015 - Jun. 2019

Bachelor of Science, Physics

Awards: 'Jingshi' Scholarship (2016 - 2018)

SKILLS

Software / Programming

- Python, MATLAB, R, SQL, LaTeX
- COMSOL, SolidWorks, Linux
- GSAS-II, Mantid, ImageJ, VESTA
- Quantum ESPRESSO, VASP
- AI Agent Training and Deployment

Materials Synthesis & Growth

- Floating-Zone Crystal Growth
- Physical Vapor Deposition / Magnetron Sputtering, Chemical Vapor Deposition

Characterization & Spectroscopy

- Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), X-ray Diffraction (XRD), Atomic Force Microscopy (AFM)
- Inelastic Neutron Scattering, Raman, Fourier Transform Infrared Spectroscopy (FTIR)
- Physical Property Measurement System (PPMS), Differential Scanning Calorimetry (DSC), Thermogravimetric Analysis (TGA)

- Four-Probe Transport Methods, Mechanical Strength Testing

Battery & Electrochemistry

- Prototype Cell Assembly, Composite Separator Fabrication
- Electrochemical Impedance Spectroscopy (EIS), Galvanostatic Cycling
- Temperature-dependent Transport Testing, Thermal Abuse Testing

Methods & Other

- Design of Experiments (DOE), Uniaxial Strain Tuning

RESEARCH EXPERIENCE

Graduate Research Assistant

University of California

Jun. 2021 - Jun. 2025

Riverside, CA

- Pioneered lattice-dynamics studies on Li-ion solid-state electrolytes, extending work previously limited to Cu- and Ag-ion systems.
- Published 5 first-author papers in journals including *PRX Energy* and *J. Mater. Chem. A*.
- Built a floating-zone crystal growth workflow from scratch, producing centimeter-scale LLZTO single crystals that enabled phonon-resolved neutron scattering measurements previously infeasible on polycrystalline samples.
- Identified diffuson-mediated thermal transport in Li-ion conductors, providing the first verification of this mechanism in Li-ion solid-state electrolytes.
- Developed a two-channel (phonon + diffuson) thermal-conductivity model in MATLAB that explained anomalous temperature dependence in Li-ion conductors where the traditional Debye model failed.

Experiment Lead

Oak Ridge National Laboratory

Jan. 2022 - Jul. 2023

Oak Ridge, TN

- Authored two successful beam time proposals and led inelastic neutron-scattering experiments on LLZTO single crystals using TAX (HFIR) and ARCS (SNS) at Oak Ridge National Laboratory (ORNL).
- Resolved the full phonon dispersion of a garnet solid electrolyte for the first time, providing experimental benchmarks for ab initio lattice-dynamics calculations.
- Built automated data-reduction pipelines in Mantid and Python, significantly reducing manual analysis time.
- Managed on-site experimental decisions and coordinated across ~7 research groups spanning UCR, UT Austin, ORNL, ETH Zürich, and TU Wien.

Student Researcher

Columbia University

Sep. 2019 - Jan. 2021

New York, NY

- Engineered composite Li-ion battery separators (γ -C₃N₄/PVDF) via solution processing, targeting improved electrochemical reaction kinetics.
- Assembled and tested CR2032 full cells over 100+ galvanostatic cycles, characterizing cycling performance and interfacial stability via EIS.
- Applied first-principles DFT (Quantum ESPRESSO) to model defect stability in olivine cathode structures, complementing experimental work with computational materials design.

Student Researcher

Beijing Normal University

Sep. 2018 - May 2019

Beijing, CN

- Designed a uniaxial-strain apparatus in SolidWorks exploiting differential thermal contraction for controlled detwinning of iron-based superconductors below 140 K.
- Built a Monte Carlo simulation in MATLAB to model atomic movement of metal atoms inside multi-wall carbon nanotubes, reproducing experimental observations.

SELECTED PUBLICATIONS

- Glass-like thermal transport in polycrystalline perovskite lithium-ion conductor $\text{Li}_{3/8}\text{Sr}_{7/16}\text{Hf}_{1/4}\text{Ta}_{3/4}\text{O}_3$
 - *Chemical Communications*, 2025 [\[Link\]](#) [\[Front Cover\]](#)
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Low Thermal Conductivity and Lattice Anharmonicity of NaSICON-type Solid Electrolyte $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$
 - *Tungsten*, 2025 [\[Link\]](#)
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte: Linking lattice and ionic dynamics with thermal transport
 - *PRX Energy*, 2025 [\[Link\]](#)
 - **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, Y. Li, H. Li, J. He, Y. Xu, S. Guo, Q. Cai, D. L. Abernathy, T. Williams, K. V. Kravchyk, M. V. Kovalenko, G. K. H. Madsen, C. Li, X. Chen
- Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$
 - *Journal of Materials Chemistry A*, 2024 [\[Link\]](#)
 - **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen
- Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $\text{Sr}_{14-x}\text{Y}_x\text{Cu}_{24}\text{O}_{41}$
 - *Journal of Applied Physics*, 2024 [\[Link\]](#)
 - S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen
- Size-dependent magnon thermal transport in a nanostructured quantum magnet
 - *Cell Reports Physical Science*, 2024 [\[Link\]](#)
 - S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen
- Crystal structure and thermoelectric properties of layered van der Waals semimetal ZrTiSe_4
 - *Chemistry of Materials*, 2022 [\[Link\]](#)
 - Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, A. A. Rezaie, V. Carta, K. N. Bozhilov, D. Luong, B. P. T. Fokwa, F. Kargar, A. A. Balandin, X. Chen
- Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors
 - *Tungsten*, 2022 [\[Link\]](#)
 - **Y. Wang**, X. Chen